Drug-discovery prep findings: licences and TB / dengue targets (2026-11-05)

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Free, read-only prep (web reads of public pages). No download of weights or datasets, no code run, no spend. These are READ-ONLY licence/availability notes from README and record pages, not a full REA; the REA (static audit of any file before it is loaded) is still required per the REA-before-adopt rule.

1 Licences and availability (read today)

Item	Finding	Open point
Boltz-2 (jwohlwend/boltz)	MIT; README: code AND weights under MIT. Affinity outputs: `affinity_probability_binary` (binder vs decoy, for hit discovery) and `affinity_pred_value` (log10 IC50, for hit-to-lead).	CORRECTION to the blueprint/ earlier chat: press said "training code included", but the README says Boltz-2 training code is "coming soon"; only Boltz-1 training docs exist. MSA server option has its own terms (not stated); cuEquivariance licence not stated. Run offline/without the public MSA server for any sensitive sequence.
DiffSBDD checkpoints (Zenodo 8183747, v1, 2023-08-09)	CC BY 4.0, 8 files, 313 MB total: crossdocked ca/fullatom x cond/ joint (17.9-77.6 MB), moad ca/ fullatom x cond/joint (16.3-44.8 MB).	Models are small (tens of MB), so full fine-tune is realistic. No per-checkpoint description on the record. Repo MIT (read earlier).
ADMET-AI	MIT; pip install admet-ai; CLI + Python; CPU ok; trained on TDC ADMET datasets.	Number of endpoints, whether model files ship in the repo, and the licences of the TDC training sets are NOT confirmed.
Chemprop	MIT (per search aggregator; LICENSE not opened)	open the LICENSE file.

2 TB targets (from a literature search; no PDB/ChEMBL query run yet)

InhA (about 80 crystal structures reported in one overview; isoniazid/ethionamide target), DprE1 (BTZ043, PBTZ169, TBA7371), gyrase GyrA/GyrB (fluoroquinolones, SPR720 on GyrB), ATP synthase (bedaquiline), QcrB (Q203), MmpL3 (a transporter structure paper, 2019). The Mdluli and Spigelman review has a table of validated targets with structure status per target. NEXT: PDB and ChEMBL counts per target (gene names inhA, dprE1, gyrA/gyrB, atpE, qcrB, mmpL3), after the robots/terms check of each API.

2.1 PDB entry counts (RCSB Search API, queried 2026-11-05)

Method: search.rcsb.org (its robots.txt returns 404 = no restriction; data.rcsb.org/rest is DISALLOWED by its robots.txt and was not used), counts only, 2 s spacing. Organism filter "Mycobacterium tuberculosis" / "Dengue virus" plus a FULL-TEXT target name. WARNING: full-text matching is loose, so multi-word names over-count (an upper bound, not a curated number); every count must be curated before use.

TB target	entries	with a ligand
InhA	131	128
DprE1	12	12
GyrB	90	87
GyrA	112	105
QcrB	4	4
KatG	28	23
RpoB	102	96
ATP synthase	417 (suspect: over-count)	398
MmpL3	1 (most MmpL3 structures may be from other mycobacteria)	1

Only InhA matches an independent figure (the overview's "about 80" is older and narrower). Counts for ATP synthase look inflated by loose matching.

3 Dengue targets

Mosnodenvir (JNJ-1802) blocks the NS3-NS4B interaction, pan-serotype (EC50 0.057-11 nM across DENV-1..4 in the Nature 2023 paper), resistance in NS4B (V91A), phase I completed per that paper; current programme status NOT checked (may have changed after 2023). Important design consequence: NS3-NS4B is a protein-protein interface, not a classic small pocket, so a pocket-conditioned generator fits it poorly. NS2B-NS3 protease and NS5 polymerase were NOT covered by the literature search. PDB counts (same method and loose-matching warning as above, entries / with ligand): NS2B-NS3 protease 47 / 29; NS5 polymerase 136 / 126; NS5 methyltransferase 135 / 126; NS3 helicase 319 / 225; NS4B 317 / 224 (suspect: NS4B and Envelope return near-identical large numbers, probably matching whole-polyprotein or antibody entries, so NS4B/Envelope are not reliable); Envelope 348 / 232. The protease (29 ligand entries) and NS5 are the realistic pocket-conditioned candidates.

4 Safety posture for these targets (stated once)

TB and dengue proteins are used only as inference-time pocket inputs; generator fine-tuning data is general protein-ligand complexes. Health generative serving still needs the governance + SciACA gated tier.